

Лабораторная работа №16

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На сервере установим fail2ban (рис. @fig-1):

```
Installing      : fail2ban-server-1.1.0-6.el9.noarch      2/5
Running scriptlet: fail2ban-server-1.1.0-6.el9.noarch      2/5
Installing      : fail2ban-firewalld-1.1.0-6.el9.noarch    3/5
Installing      : fail2ban-sendmail-1.1.0-6.el9.noarch    4/5
Installing      : fail2ban-1.1.0-6.el9.noarch             5/5
Running scriptlet: fail2ban-selinux-1.1.0-6.el9.noarch    5/5
Running scriptlet: fail2ban-1.1.0-6.el9.noarch            5/5
Verifying       : fail2ban-1.1.0-6.el9.noarch             1/5
Verifying       : fail2ban-firewalld-1.1.0-6.el9.noarch    2/5
Verifying       : fail2ban-selinux-1.1.0-6.el9.noarch    3/5
Verifying       : fail2ban-sendmail-1.1.0-6.el9.noarch    4/5
Verifying       : fail2ban-server-1.1.0-6.el9.noarch      5/5

Installed:
fail2ban-1.1.0-6.el9.noarch          fail2ban-firewalld-1.1.0-6.el9.noarch
fail2ban-selinux-1.1.0-6.el9.noarch  fail2ban-sendmail-1.1.0-6.el9.noarch
fail2ban-server-1.1.0-6.el9.noarch

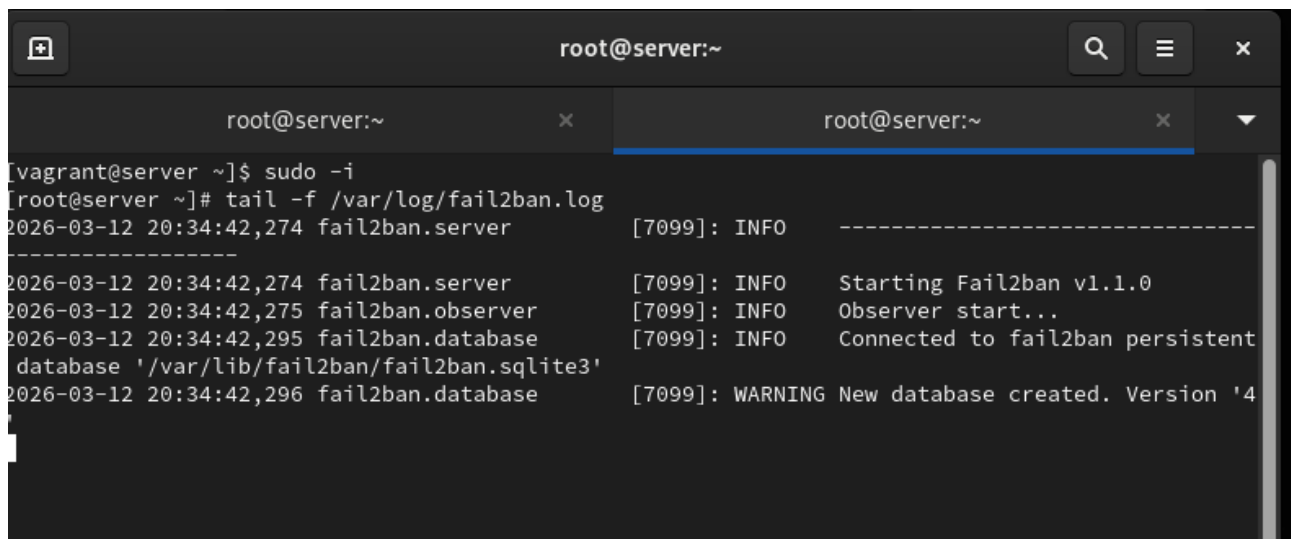
Complete!
[root@server ~]# systemctl start fail2ban
[root@server ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
[root@server ~]# S
```

Рис. 1: Этап 1

Запустим сервер fail2ban и добавим его в автозагрузку (рис. @fig-2):

В дополнительном терминале запустим просмотр журнала событий fail2ban (рис. @fig-3):

Создадим файл с локальной конфигурацией fail2ban (рис. @fig-4):



A terminal window titled 'root@server:~' showing the output of the command 'tail -f /var/log/fail2ban.log'. The log shows fail2ban starting, connecting to a persistent database, and creating a new database. The user is root@server.

```
root@server:~  
[vagrant@server ~]$ sudo -i  
[root@server ~]# tail -f /var/log/fail2ban.log  
2026-03-12 20:34:42,274 fail2ban.server [7099]: INFO -----  
-----  
2026-03-12 20:34:42,274 fail2ban.server [7099]: INFO Starting Fail2ban v1.1.0  
2026-03-12 20:34:42,275 fail2ban.observer [7099]: INFO Observer start...  
2026-03-12 20:34:42,295 fail2ban.database [7099]: INFO Connected to fail2ban persistent  
database '/var/lib/fail2ban/fail2ban.sqlite3'  
2026-03-12 20:34:42,296 fail2ban.database [7099]: WARNING New database created. Version '4'
```

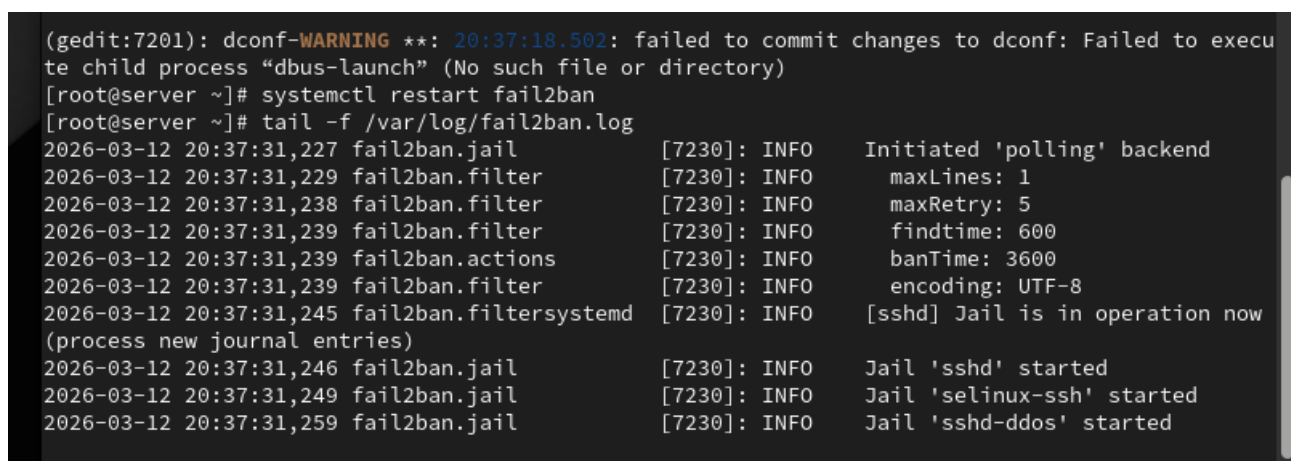
Рис. 2: Этап 2



A text editor window titled 'customisation.local' showing the configuration file '/etc/fail2ban/jail.d'. The file contains configuration for the default jail, including bantime, filter, and enabled status for various services like sshd, selinux-ssh, and sshd-ddos. The user is root@server.

```
Open customisation.local /etc/fail2ban/jail.d Save  
1 [DEFAULT]  
2 bantime = 3600  
3  
4 #  
5 # SSH servers  
6 #  
7 [sshd]  
8 port = ssh,2022  
9 enabled = true  
10 [sshd-ddos]  
11 filter = sshd  
12 enabled = true  
13 [selinux-ssh]  
14 enabled = true
```

Рис. 3: Этап 3



A terminal window showing the restart of fail2ban and its log output. The log shows fail2ban initiating the 'polling' backend, starting the jail 'sshd', and starting the jail 'selinux-ssh'. The user is root@server.

```
(gedit:7201): dconf-WARNING **: 20:37:18.502: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart fail2ban  
[root@server ~]# tail -f /var/log/fail2ban.log  
2026-03-12 20:37:31,227 fail2ban.jail [7230]: INFO Initiated 'polling' backend  
2026-03-12 20:37:31,229 fail2ban.filter [7230]: INFO maxLines: 1  
2026-03-12 20:37:31,238 fail2ban.filter [7230]: INFO maxRetry: 5  
2026-03-12 20:37:31,239 fail2ban.filter [7230]: INFO findtime: 600  
2026-03-12 20:37:31,239 fail2ban.actions [7230]: INFO banTime: 3600  
2026-03-12 20:37:31,239 fail2ban.filter [7230]: INFO encoding: UTF-8  
2026-03-12 20:37:31,245 fail2ban.filtersystemd [7230]: INFO [sshd] Jail is in operation now  
(process new journal entries)  
2026-03-12 20:37:31,246 fail2ban.jail [7230]: INFO Jail 'sshd' started  
2026-03-12 20:37:31,249 fail2ban.jail [7230]: INFO Jail 'selinux-ssh' started  
2026-03-12 20:37:31,259 fail2ban.jail [7230]: INFO Jail 'sshd-ddos' started
```

Рис. 4: Этап 4

В файле /etc/fail2ban/jail.d/customisation.local зададим время блокирования на 1 час и включим защиту SSH (рис. @fig-5):

```
database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:37:31,146 fail2ban.jail [7230]: INFO Creating new jail 'sshd'
2026-03-12 20:37:31,150 fail2ban.jail [7230]: INFO Jail 'sshd' uses systemd {}
2026-03-12 20:37:31,151 fail2ban.jail [7230]: INFO Initiated 'systemd' backend
2026-03-12 20:37:31,156 fail2ban.filter [7230]: INFO maxLines: 1
2026-03-12 20:37:31,180 fail2ban.filtersystemd [7230]: INFO [sshd] Added journal match for:
'_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session'
2026-03-12 20:37:31,180 fail2ban.filter [7230]: INFO maxRetry: 5
2026-03-12 20:37:31,180 fail2ban.filter [7230]: INFO findtime: 600
2026-03-12 20:37:31,180 fail2ban.actions [7230]: INFO banTime: 3600
2026-03-12 20:37:31,180 fail2ban.filter [7230]: INFO encoding: UTF-8
2026-03-12 20:37:31,181 fail2ban.jail [7230]: INFO Creating new jail 'selinux-ssh'
2026-03-12 20:37:31,182 fail2ban.jail [7230]: INFO Jail 'selinux-ssh' uses poller {}
2026-03-12 20:37:31,182 fail2ban.jail [7230]: INFO Initiated 'polling' backend
2026-03-12 20:37:31,183 fail2ban.datedetector [7230]: INFO date pattern `''`: `Epoch`
2026-03-12 20:37:31,184 fail2ban.filter [7230]: INFO maxRetry: 5
2026-03-12 20:37:31,184 fail2ban.filter [7230]: INFO findtime: 600
2026-03-12 20:37:31,184 fail2ban.actions [7230]: INFO banTime: 3600
2026-03-12 20:37:31,184 fail2ban.filter [7230]: INFO encoding: UTF-8
2026-03-12 20:37:31,224 fail2ban.filter [7230]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 0, hash = 0c2be31ffde8214ab616b0e504468ebe00b577f7)
2026-03-12 20:37:31,224 fail2ban.jail [7230]: INFO Creating new jail 'sshd-ddos'
2026-03-12 20:37:31,226 fail2ban.jail [7230]: INFO Jail 'sshd-ddos' uses poller {}
2026-03-12 20:37:31,227 fail2ban.jail [7230]: INFO Initiated 'polling' backend
2026-03-12 20:37:31,229 fail2ban.filter [7230]: INFO maxLines: 1
2026-03-12 20:37:31,238 fail2ban.filter [7230]: INFO maxRetry: 5
2026-03-12 20:37:31,239 fail2ban.filter [7230]: INFO findtime: 600
2026-03-12 20:37:31,239 fail2ban.actions [7230]: INFO banTime: 3600
2026-03-12 20:37:31,239 fail2ban.filter [7230]: INFO encoding: UTF-8
2026-03-12 20:37:31,245 fail2ban.filtersystemd [7230]: INFO [sshd] Jail is in operation now
(process new journal entries)
2026-03-12 20:37:31,246 fail2ban.jail [7230]: INFO Jail 'sshd' started
2026-03-12 20:37:31,249 fail2ban.jail [7230]: INFO Jail 'selinux-ssh' started
2026-03-12 20:37:31,259 fail2ban.jail [7230]: INFO Jail 'sshd-ddos' started
```

Рис. 5: Этап 5

Перезапустим fail2ban для применения изменений (рис. @fig-6):

Посмотрим журнал событий после перезапуска (рис. @fig-7):

В файле /etc/fail2ban/jail.d/customisation.local включим защиту HTTP (рис. @fig-8):

Перезапустим fail2ban и проверим журнал событий (рис. @fig-9, @fig-10):

Перезапуск и проверка

```
1 [DEFAULT]
2 bantime = 3600
3
4 #
5 # SSH servers
6 #
7 [sshd]
8 port = ssh,2022
9 enabled = true
10 [sshd-ddos]
11 filter = sshd
12 enabled = true
13 [selinux-ssh]
14 enabled = true
15
16 #
17 # HTTP servers
18 #
19 [apache-auth]
20 enabled = true
21 [apache-badbots]
22 enabled = true
23 [apache-noscript]
24 enabled = true
25 [apache-overflows]
26 enabled = true
27 [apache-nohome]
28 enabled = true
29 [apache-botsearch]
30 enabled = true
31 [apache-fakegooglebot]
32 enabled = true
33 [apache-modsecurity]
34 enabled = true
35
36 [apache-shellshock]
37 enabled = true
```

Saving file "/etc/fail2ban/jail.d/customisation.local"... Plain Text ▾ Tab Width: 8 ▾ Ln 27, Col 16 ▾ INS

Рис. 6: Этап 6

```
root@server:~  
root@server:~ x root@server:~ x root@server:~ x  
** (gedit:7296): WARNING **: 20:39:15.499: Set document metadata failed: Setting attribute metadata::gedit-position not supported  
(gedit:7296): dconf-WARNING **: 20:39:15.517: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart fail2ban  
[root@server ~]# tail -f /var/log/fail2ban.log  
2026-03-12 20:39:27,417 fail2ban.filter [7328]: WARNING [apache-botsearch] Please check a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) - may result in undefined or erroneous behavior  
2026-03-12 20:39:27,431 fail2ban.jail [7328]: INFO Jail 'apache-botsearch' started  
2026-03-12 20:39:27,440 fail2ban.jail [7328]: INFO Jail 'apache-fakegooglebot' started  
2026-03-12 20:39:27,448 fail2ban.filter [7328]: WARNING [apache-modsecurity] Ignoring all log entries older than 600s; these are probably messages generated while fail2ban was not running.  
2026-03-12 20:39:27,448 fail2ban.filter [7328]: WARNING [apache-modsecurity] Please check a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) - may result in undefined or erroneous behavior  
2026-03-12 20:39:27,449 fail2ban.jail [7328]: INFO Jail 'apache-modsecurity' started  
2026-03-12 20:39:27,461 fail2ban.filter [7328]: WARNING [apache-shellshock] Ignoring all log entries older than 600s; these are probably messages generated while fail2ban was not running.  
2026-03-12 20:39:27,470 fail2ban.jail [7328]: INFO Jail 'apache-shellshock' started  
2026-03-12 20:39:27,470 fail2ban.filter [7328]: WARNING [apache-shellshock] Please check a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) - may result in undefined or erroneous behavior  
2026-03-12 20:39:27,508 fail2ban.jail [7328]: INFO Jail 'sshd-ddos' started
```

Рис. 7: Этап 7

```

Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
  |- Currently banned: 0
  |- Total banned: 0
  `-- Banned IP list:
[root@server ~]# fail2ban-client set sshd maxretry 2
2
[root@server ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
  |- Currently banned: 0
  |- Total banned: 0
  `-- Banned IP list:
[root@server ~]# fail2ban-client set sshd unbanip <ip-адрес клиента>
-bash: syntax error near unexpected token `newline'
[root@server ~]# fail2ban-client set sshd unbanip 128.45.29.1.1
0
[root@server ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
  |- Currently banned: 0
  |- Total banned: 0
  `-- Banned IP list:
[root@server ~]#

```

Рис. 8: Этап 8



```

*customisation.local
/etc/fail2ban/jail.d

1 [DEFAULT]
2 bantime = 3600
3 ignoreip = 127.0.0.1/8
4 #
5 # SSH servers
6 #
7 [sshd]
8 port = ssh,2022
9 enabled = true

```

Рис. 9: Этап 9

```

** (gedit:7438): WARNING **: 20:44:29.558: Set document metadata failed: Setting attribute metadata::gedit-position not supported

(gedit:7438): dconf-WARNING **: 20:44:29.584: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:44:38,221 fail2ban.jail [7469]: INFO Jail 'apache-auth' started
2026-03-12 20:44:38,247 fail2ban.jail [7469]: INFO Jail 'apache-badbots' started
2026-03-12 20:44:38,251 fail2ban.jail [7469]: INFO Jail 'apache-noscript' started
2026-03-12 20:44:38,253 fail2ban.jail [7469]: INFO Jail 'apache-overflows' started
2026-03-12 20:44:38,257 fail2ban.jail [7469]: INFO Jail 'apache-nohome' started
2026-03-12 20:44:38,294 fail2ban.jail [7469]: INFO Jail 'apache-botsearch' started
2026-03-12 20:44:38,317 fail2ban.jail [7469]: INFO Jail 'apache-fakegooglebot' started
2026-03-12 20:44:38,360 fail2ban.jail [7469]: INFO Jail 'apache-modsecurity' started
2026-03-12 20:44:38,392 fail2ban.jail [7469]: INFO Jail 'apache-shellshock' started
2026-03-12 20:44:38,402 fail2ban.jail [7469]: INFO Jail 'sshd-ddos' started

```

Рис. 10: Этап 10

```

root@server:~ x root@server:/vagrant/provision/server x
[vagrant@server ~]$ sudo -i
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d cp -R /etc/fail2ban/jail.d/customisation.local
mkdir: invalid option -- 'R'
Try 'mkdir --help' for more information.
[root@server server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d
[root@server server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provision/server/protect/etc/fail2ban/jail.d/
[root@server server]# gedit /vagrant/provision/server

(gedit:7569): dconf-WARNING **: 20:46:46.692: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

```

Рис. 11: Этап 11

В файле `/etc/fail2ban/jail.d/customisation.local` включим защиту почты (рис. @fig-11):

Перезапустим `fail2ban` и посмотрим журнал событий (рис. @fig-12, @fig-13):



```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install fail2ban
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/server/protect/etc/* /etc
7 restorecon -vR /etc
8 echo "Start fail2ban service"
9 systemctl enable fail2ban
10 systemctl start fail2ban
```

Рис. 12: Этап 12

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?